

Logistic regression results

For a little background, I first tried to build a Neural network incorporating transfer learning to it. But nothing seemed to work with the data, I kept getting the same errors. Transfer learning is more used for image and audio data rather than tabular data like ours. So I decided to move on to logistic regression.

The first simple logistic regression model:

```
Overall Model Performance:
Overall Accuracy: 0.5838667366211963
Macro-Averaged Precision: 0.5882673553987926
Macro-Averaged Recall: 0.5845534046389963
Macro-Averaged F1-score: 0.5772422158075537
```

The second logistic regression model with grid search:

```
Classification Report (Regularized Logistic Regression):
              precision    recall  f1-score   support

      High         0.64         0.61         0.62        12607
       Low         0.55         0.73         0.63        12685
      Medium        0.58         0.41         0.48        12828

 accuracy          0.58          0.58          0.58        38120
 macro avg         0.59          0.58          0.58        38120
 weighted avg      0.59          0.58          0.58        38120

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```

The third model, now using random forest classifier. I changed to this model to see if I would still get low scores.

```
Best Parameters for Random Forest: {'max_depth': 30, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 300}
Validation Accuracy (RF): 0.9472
Validation Precision (RF): 0.9482
Validation Recall (RF): 0.9472
Validation F1-Score (RF): 0.9471
Classification Report (RF):
              precision    recall  f1-score   support

      High         0.93         1.00         0.96        10088
       Low         0.98         0.93         0.95        10330
      Medium        0.93         0.92         0.92        10078

 accuracy          0.95          0.95          0.95        30496
 macro avg         0.95          0.95          0.95        30496
 weighted avg      0.95          0.95          0.95        30496
```

By using these models, I have found out that Logistic regression and transfer learning are not the best to use for our type of data. From the results, it is clear to see that it is best to choose a model such as Random Forest Classifier, which yields a 95% accuracy.